Paper Review • arXiv 2611.06817

Open-HumanWAM

An open recipe for scaling embodied foundation models with 50k hours of egocentric human video

An Open Recipe for Scaling Human and Robot Experience for Embodied Foundation Models

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Note: the prompt title is “Open-HumanScale”; the paper text and open-source release use Open-HumanWAM

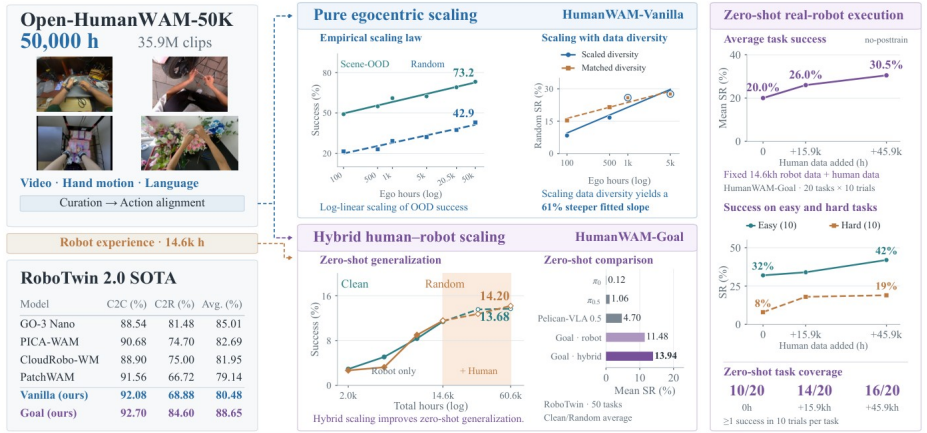


Image: paper Figure 1 (overview)

ONE-SENTENCE SUMMARY

More human video → better generalization; +robot data → zero-shot

After pretraining a world–action model on **50k hours** of open egocentric human video (hand actions + language labels), robot **OOD success rises roughly log-linearly with pretraining hours**; adding 14.6k h of robot data and goal-image task specification, HumanWAM-Goal reaches **88.65% avg. success (SOTA)** on RoboTwin 2.0, executes zero-shot, and changes behavior when only the goal image changes.

50,000 h

Open-HumanWAM-50K: 5 sources, 35.9M clips, unified 16-D bimanual actions

49.1→73.2%

IDM Scene-OOD success, 100 h → 50k h human pretraining (R^2=0.969)

88.65%

HumanWAM-Goal RoboTwin 2.0 avg. (Clean 92.70 / Random 84.60)

20.0→30.5%

Real-robot zero-shot avg. success on 20 tasks, +45.9k h human data

Three-part storyline: ① build and release the corpus → ② scaling study of human-only pretraining (HumanWAM-Vanilla) → ③ mixed human + robot training with goal-image conditioning (HumanWAM-Goal) → real-robot zero-shot, in-context learning and long-horizon tasks

MOTIVATION

Prior work shows “human data helps”, but mostly via loss and without open release

Prior work

EgoScale

20k+ hours of action-labeled egocentric pretraining; data scale vs. validation loss is log-linear

HumanScale

Post-trained val. action loss falls log-linearly with human hours; human > equal-hour robot data

Dyna-2

Scales to 1M hours; reports scaling of zero-shot action prediction

Gaps: mostly loss / open-loop metrics; ID and OOD conflated; data and pipelines mostly unreleased.

Questions addressed

Scale — 100→50k h: do closed-loop gains come in-distribution or OOD?

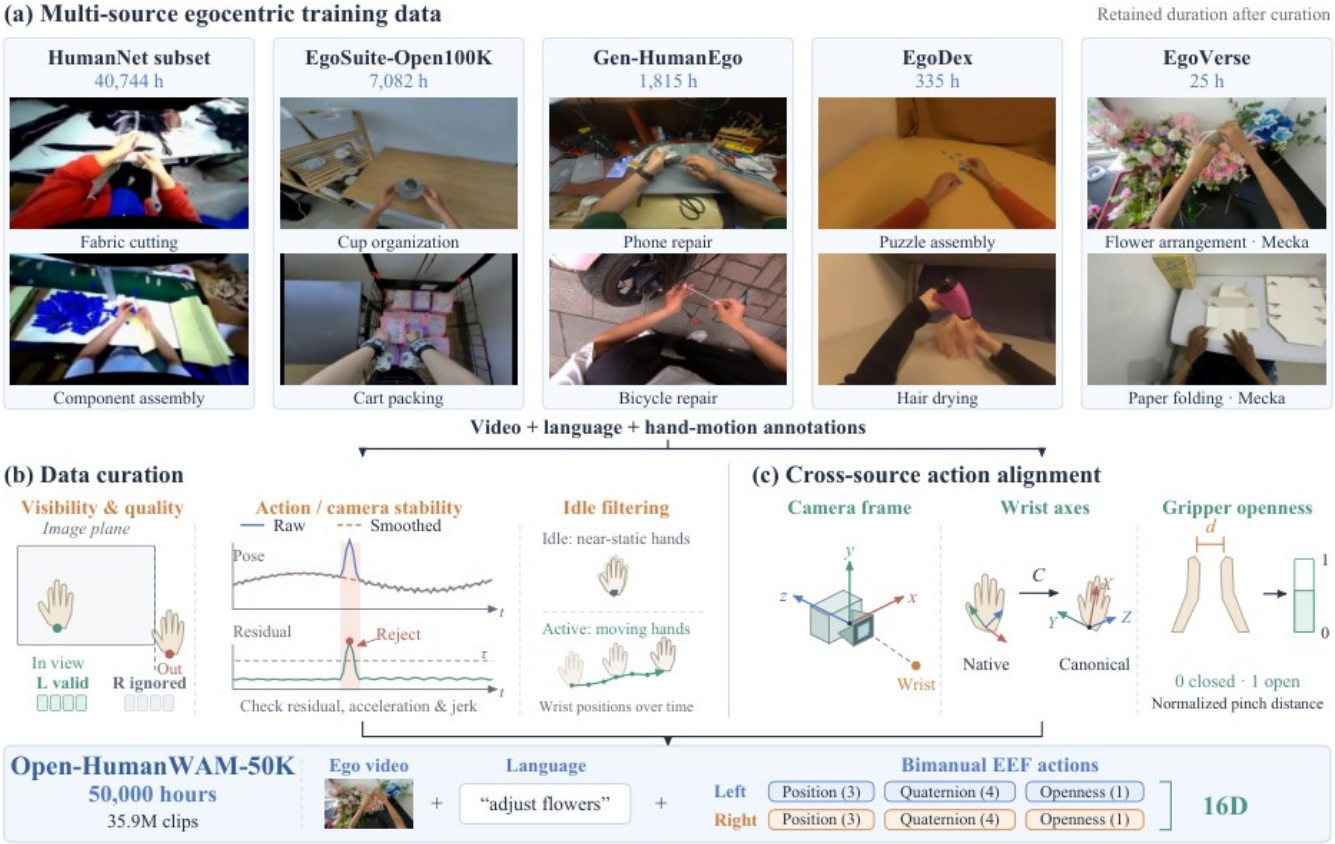
Recipe — Roles of diversity, cross-source action alignment and human action supervision?

Source — At equal hours, can human data beat robot data?

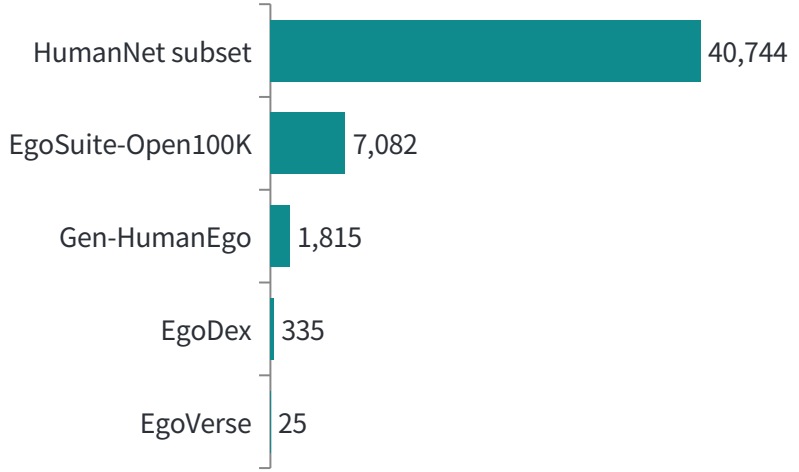
Paradigm — Can mixed human+robot data enable zero-shot use without per-task fine-tuning?

OPEN-HUMANWAM-50K

Data: 104k hours of raw video, 50k hours kept after cleaning and alignment



Cleaned hours per source



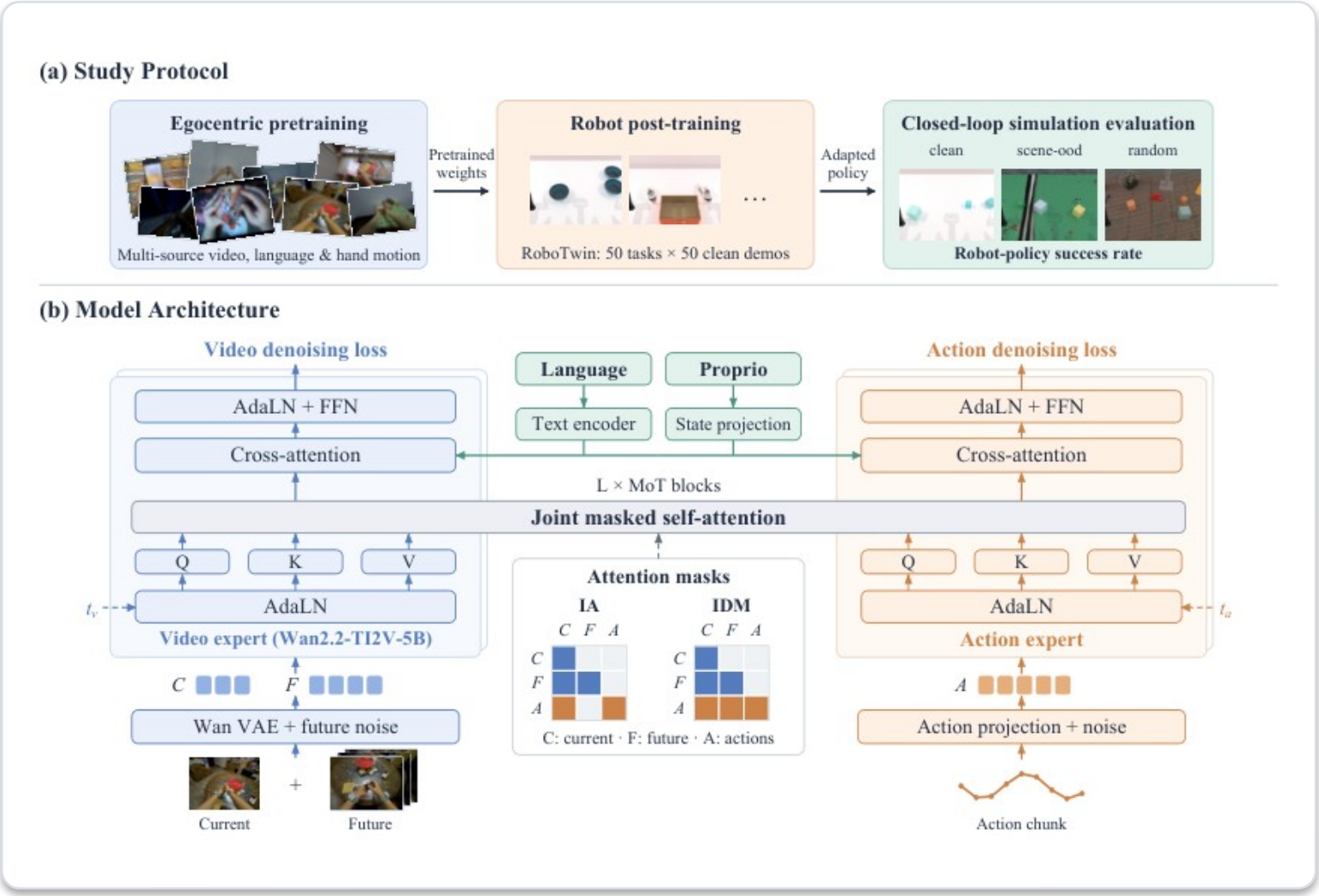
104k → 50k h — Before → after cleaning

35.9M — Continuous clips (31.4M raw episodes)

16-D — Bimanual EEF: pos. 3 + quat. 4 + gripper 1

MODEL & PROTOCOL

HumanWAM-Vanilla: video+action experts, joint shared-attn. denoising



Backbone

Wan2.2-TI2V-5B video expert + equal-depth action expert (MoT)

Two masks

IA: actions see current frame only (fast)
IDM: actions also see future video frames

Final config

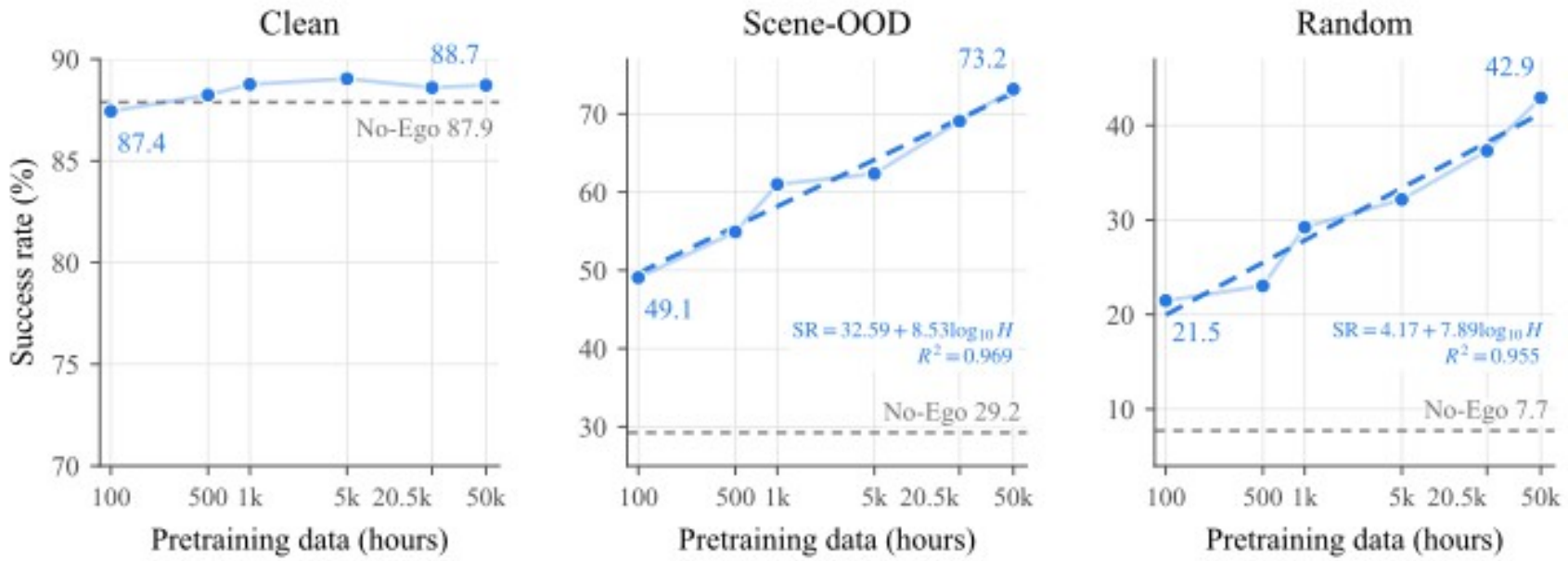
IDM + camera-frame absolute poses; 32-step action chunks, future video generated before actions

Three eval tiers

clean (in-distribution) / scene-ood (new background textures) / random (clutter, lighting, table height)

Q1 • SCALING 100 → 50,000 HOURS

Scale mainly buys OOD generalization: OOD success rises log-linearly with hours



Top: IDM attention, 100 h → 50k h (gray dashed = no-human-pretraining baseline)

87.4 → 88.7%

Clean: roughly flat (No-Ego 87.9%)

49.1 → 73.2%

Scene-OOD： R^2=0.969 （No-Ego 29.2%）

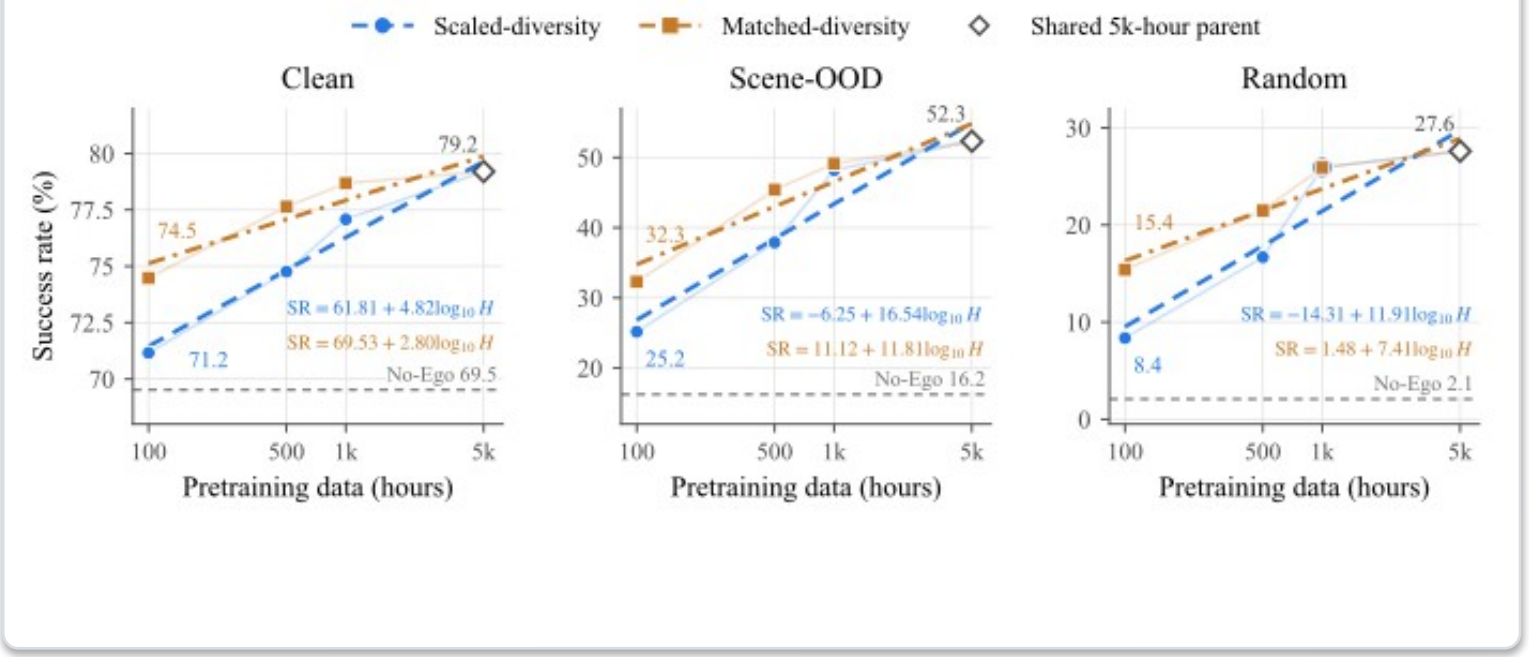
21.5 → 42.9%

Random： R^2=0.955 （No-Ego 7.7%）

+4.4/+15.9/+10.2

IA variant gain per 10× data (pp): Clean / Scene-OOD / Random

Diversity is also a scaling axis: at equal hours, broader task/scene coverage is better



At 100 h	Tasks	Scenes	Clean	Scene-OOD	Random
Scaled-diversity (blue)	62	222	71.2	25.2	8.4
Matched-diversity (orange)	**2,288**	**2,953**	**74.5**	**32.3**	**15.4**

Paper's claim

Task/scene coverage growing with scale yields a steeper fitted slope:

Clean +2.0 • Scene-OOD +4.7 • Random +4.5

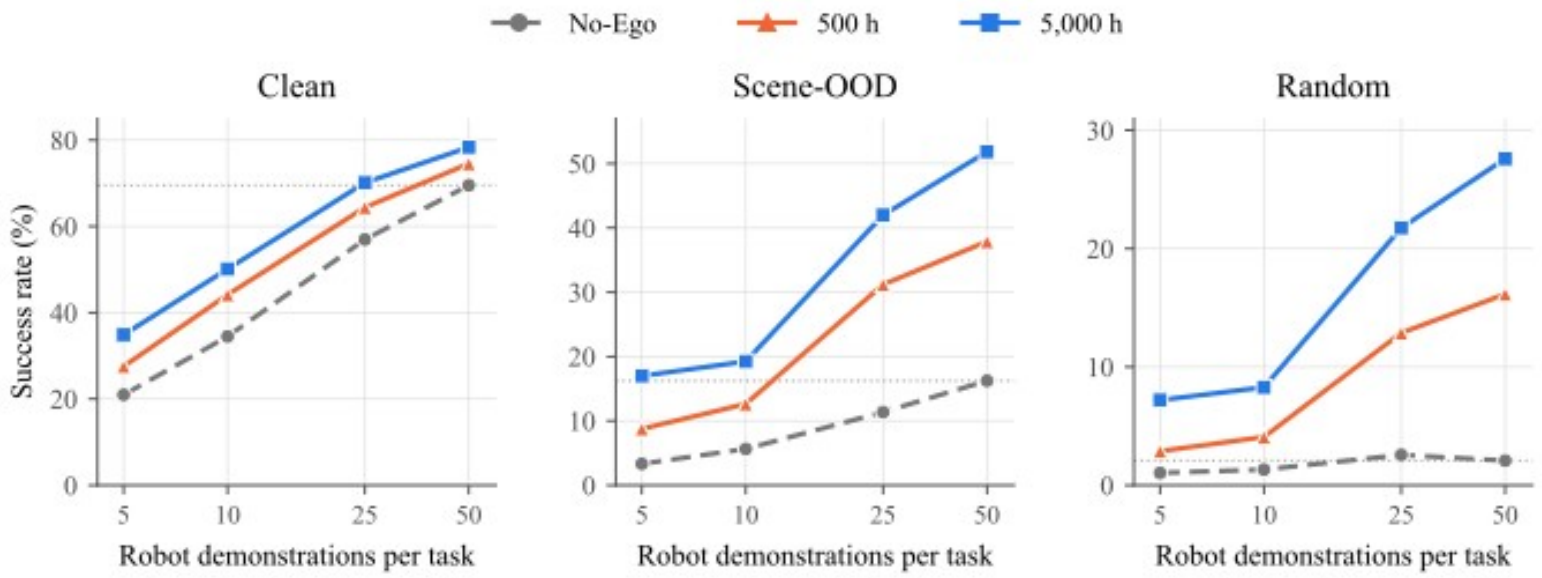
Caveats

Both groups share the same parent set at 5k h, so endpoints must coincide;

at 100–1k h, the broader-coverage matched group is actually higher.

→ More direct conclusion: at equal hours, broader coverage is better.

More pretraining = fewer robot demos: half the demos, better generalization



34.5 → 44.2 → 50.1%

Clean success with 10 demos per task
No-Ego → 500 h → 5,000 h

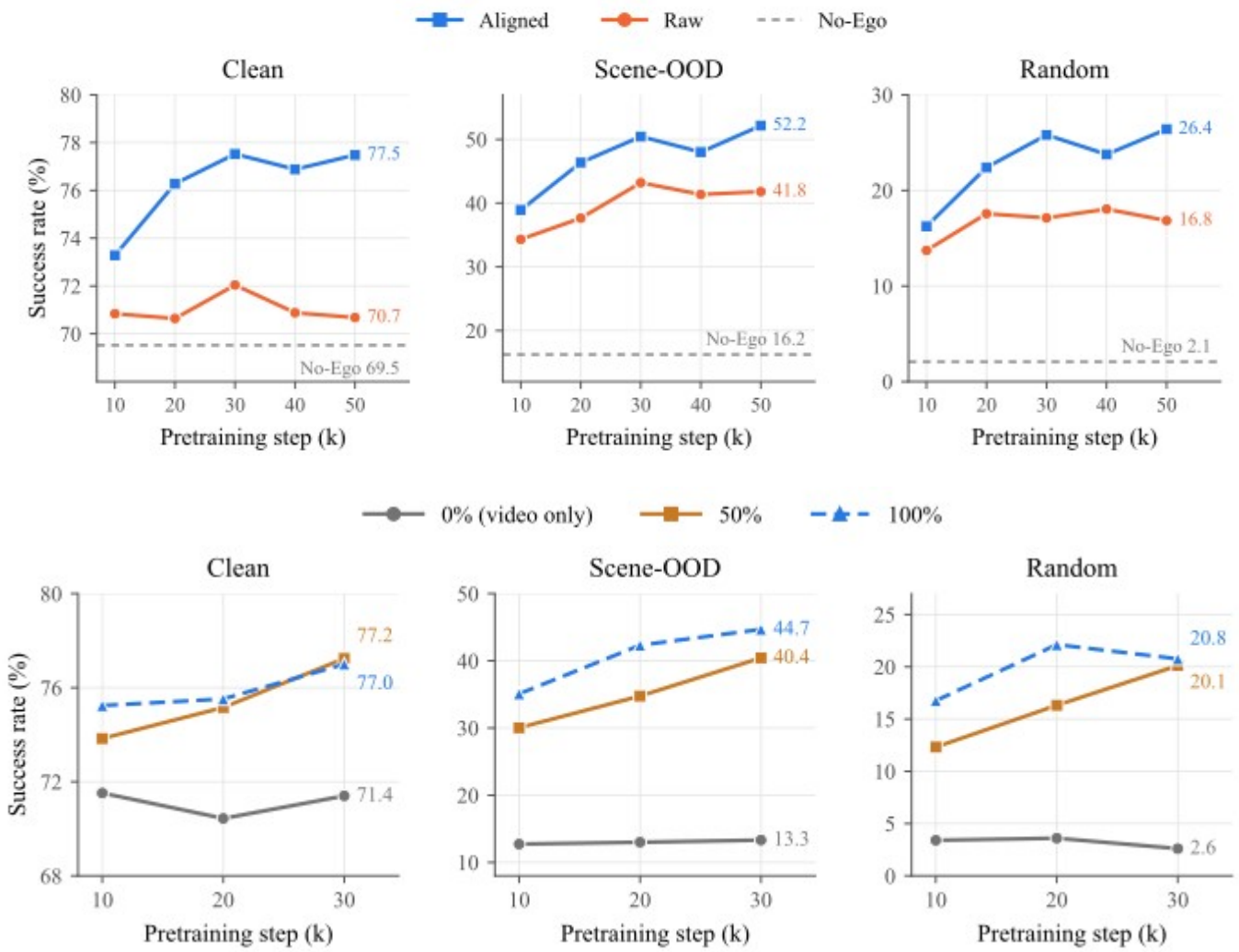
25 demos > 50 demos

5k h + 25 demos beats 500 h + 50 demos:
Scene-OOD 42.0 vs 37.9 • Random 21.8 vs 16.2

5 demos > 50 demos

Scene-OOD: 5k h model with 5 demos beats
No-Ego with 50 demos

Two musts: cross-source alignment + human actions; video-only ≈ useless



Cross-source alignment (1k h, 4 sources)

Scene-OOD 52.2 vs 41.8

Random 26.4 vs 16.8

Alignment lowers action loss with nearly identical video loss (Fig.10): it makes multi-source action supervision more consistent.

Human action supervision

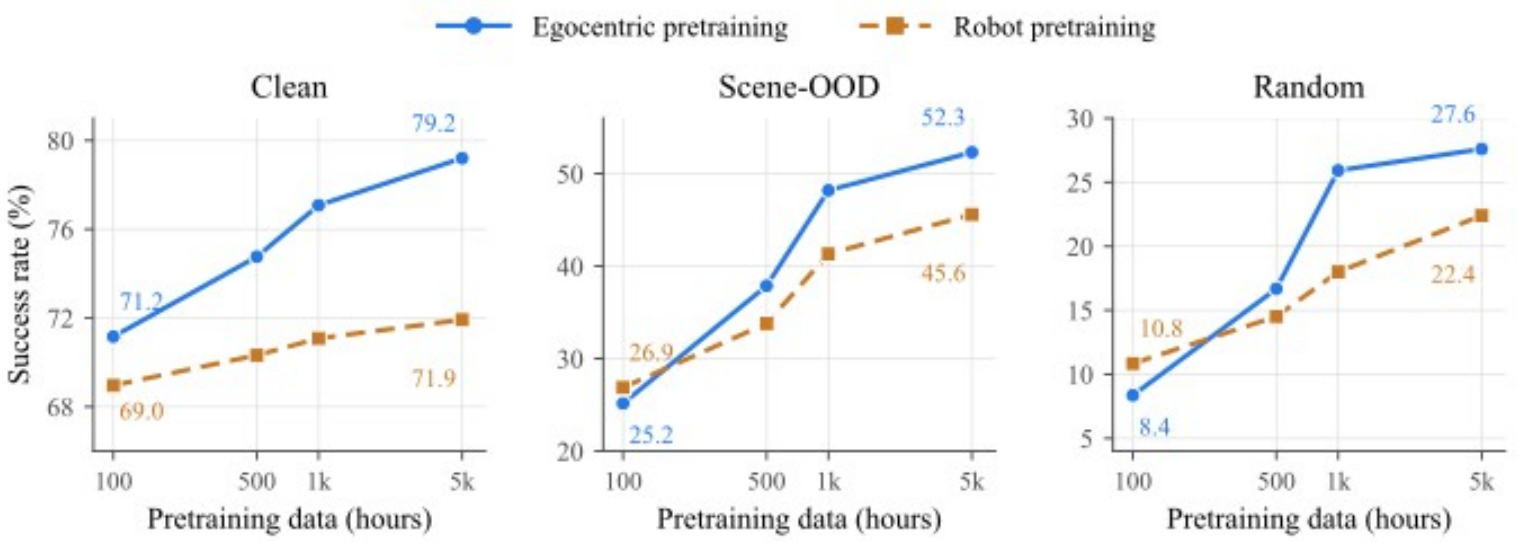
Video prediction only: Scene-OOD ≈13%, Random ≈3%, no gain with longer training

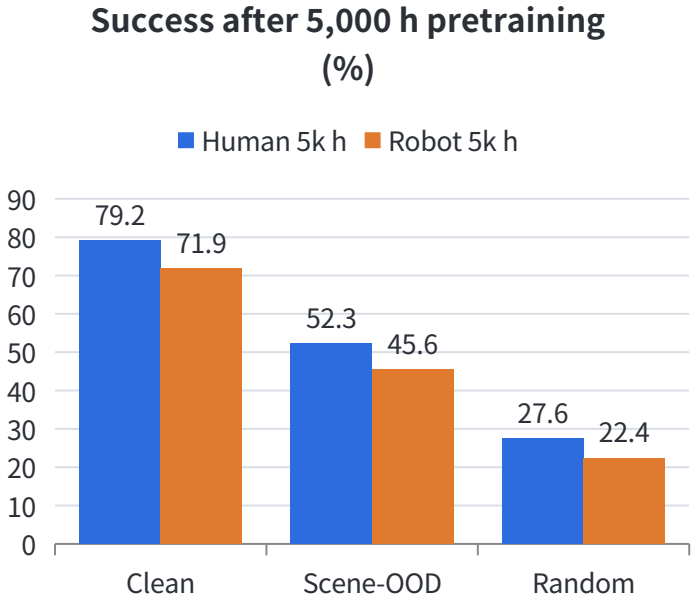
With action supervision (30k steps): Scene-OOD 40.4% (50%) / 44.7% (100%)

50% and 100% label ratios perform similarly.

Source: paper Figure 9, Figure 11

At equal hours, human video beats robot data (better across the board at ⩾500 h)





1k h human > 5k h robot

Scene-OOD 48.2 vs 45.6 • Random 25.9 vs 22.4

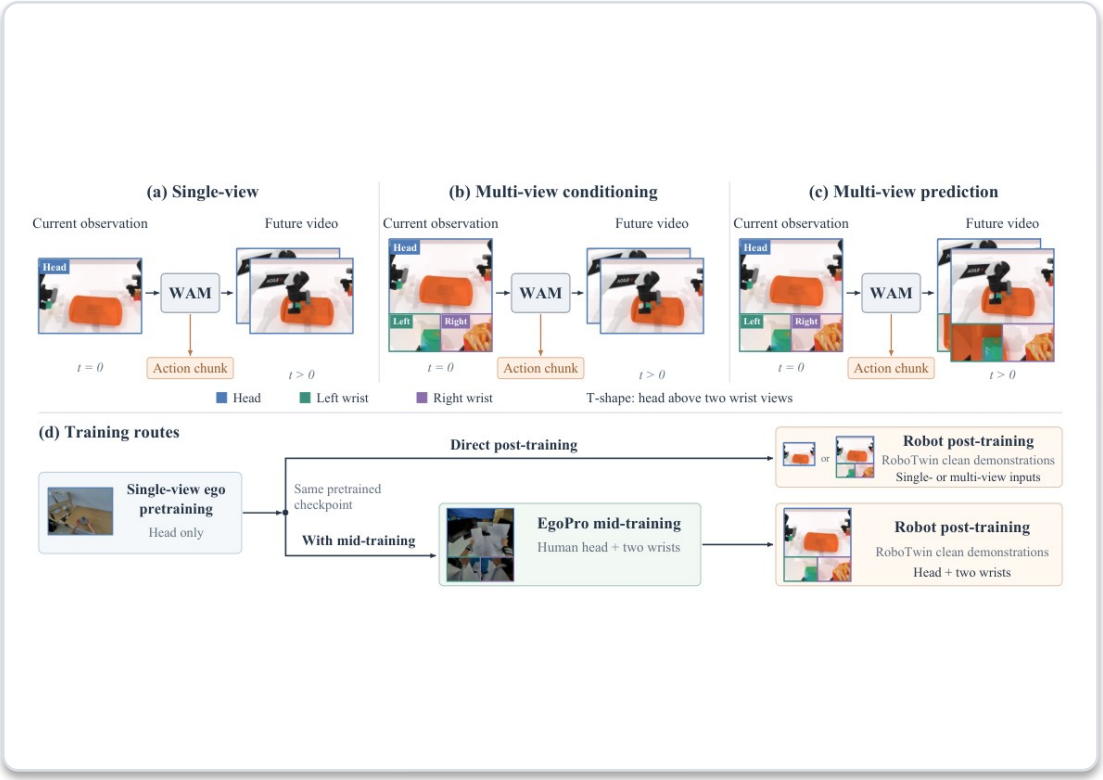
Robot slightly better at 100 h

Scene-OOD 26.9 vs 25.2 • Random 10.8 vs 8.4

Clean gain 2.2→7.3 pp

Robot pretraining: Clean barely improves with scale

Wrist cams dilute human priors; multi-view human mid-training fixes it



Config (RoboTwin success %)	Mid-train	Clean	Scene-OOD	Random
IA • single-view	—	75.68	46.08	22.48
IA • multi-view cond.	—	74.60	34.08	12.92
IA • multi-view pred.	—	75.04	33.00	14.84
IDM • single-view	—	88.48	62.00	28.56
IDM • multi-view cond.	—	90.56	59.52	29.08
IDM • multi-view cond.	**EgoPro**	**90.28**	**61.20**	**30.68**
IDM • multi-view pred.	—	89.84	56.28	26.16
IDM • multi-view pred.	EgoPro	88.72	58.80	27.36

Conclusion

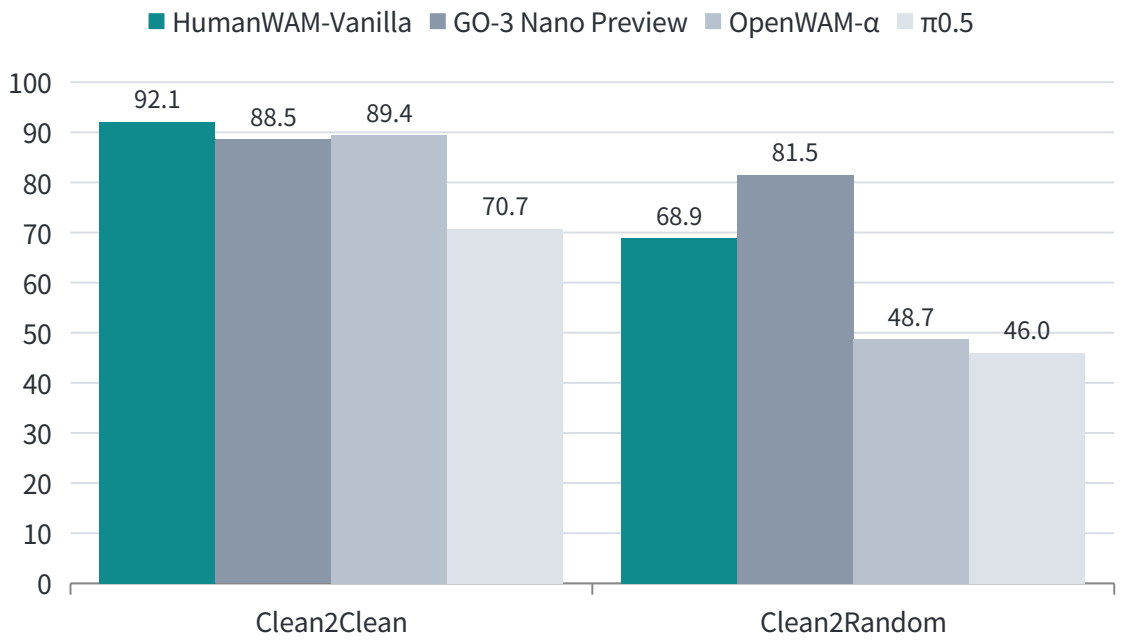
Adding wrist views directly under IA: Scene-OOD 46.1 → 33.0–34.1%; **IDM + EgoPro mid-training + multi-view conditioning**: Scene-OOD 61.2% (near single-view 62.0), Random 30.7% (above 28.6) → chosen as the final route

Source: paper Figure 13, Table 5

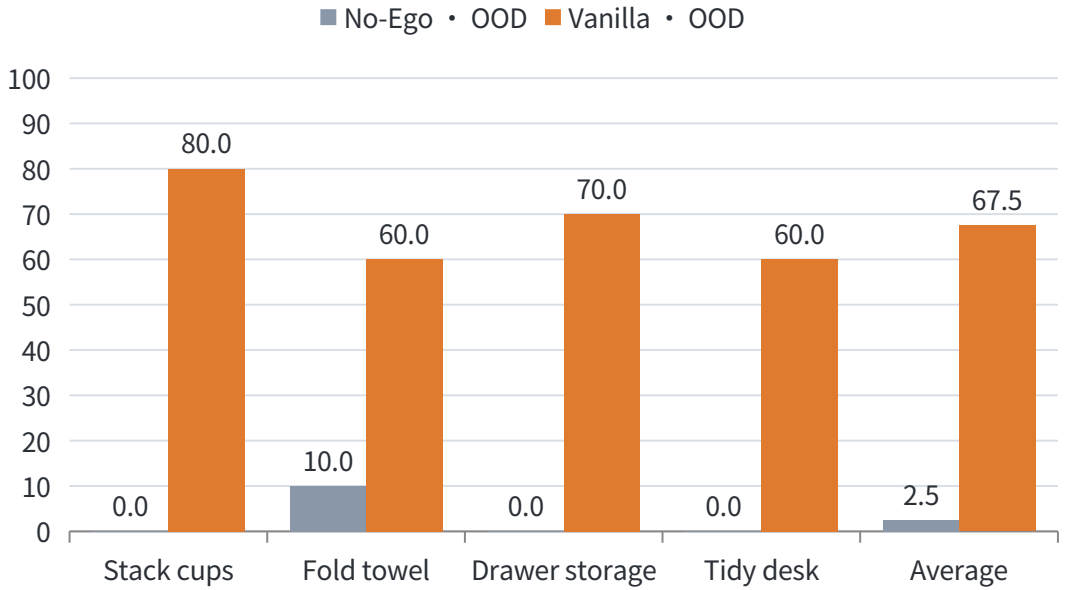
HUMANWAM-VANILLA RESULTS

HumanWAM-Vanilla: #1 on Clean2Clean; real-robot OOD 2.5% → 67.5%

RoboTwin 2.0 (post-trained on clean demos only, %)



Real Cobot Magic: success after tablecloth/lighting changes (%, 10 trials each)



92.08%

RoboTwin Clean2Clean, best in table; avg. 80.48%

14.17% / 19.00

RoboDojo avg. success / progress score; Precision 20.5%, Long-Horizon 33.0%

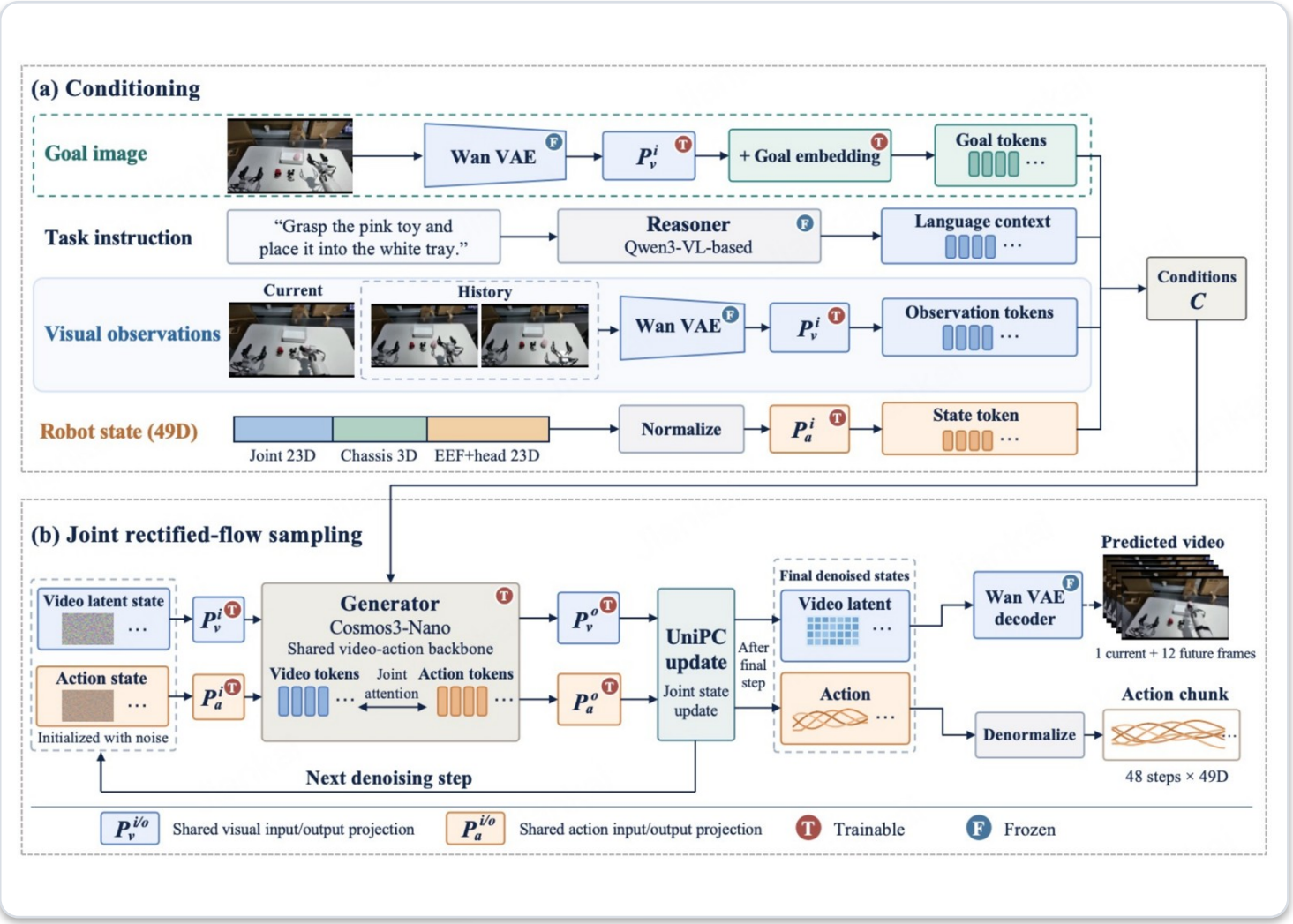
57.5 → 87.5%

Real-robot in-distribution avg. success (No-Ego → Vanilla)

Source: paper Table 7, Table 8, Table 9

HYBRID HUMAN–ROBOT MODEL

HumanWAM-Goal: goal-image tasks, 49-D actions for humans & robots



Backbone

Cosmos3-Nano-16B: frozen Qwen3-VL Reasoner + shared video-action Generator

49-D actions

Joint 23 + base 3 + EEF • head 23; per-embodiment joint proj., dropout

Cross-embodiment key

In-chunk relative EEF motion + gripper in m (÷0.12 m) → norm. stats transfer to new robots

Goal-image sampling

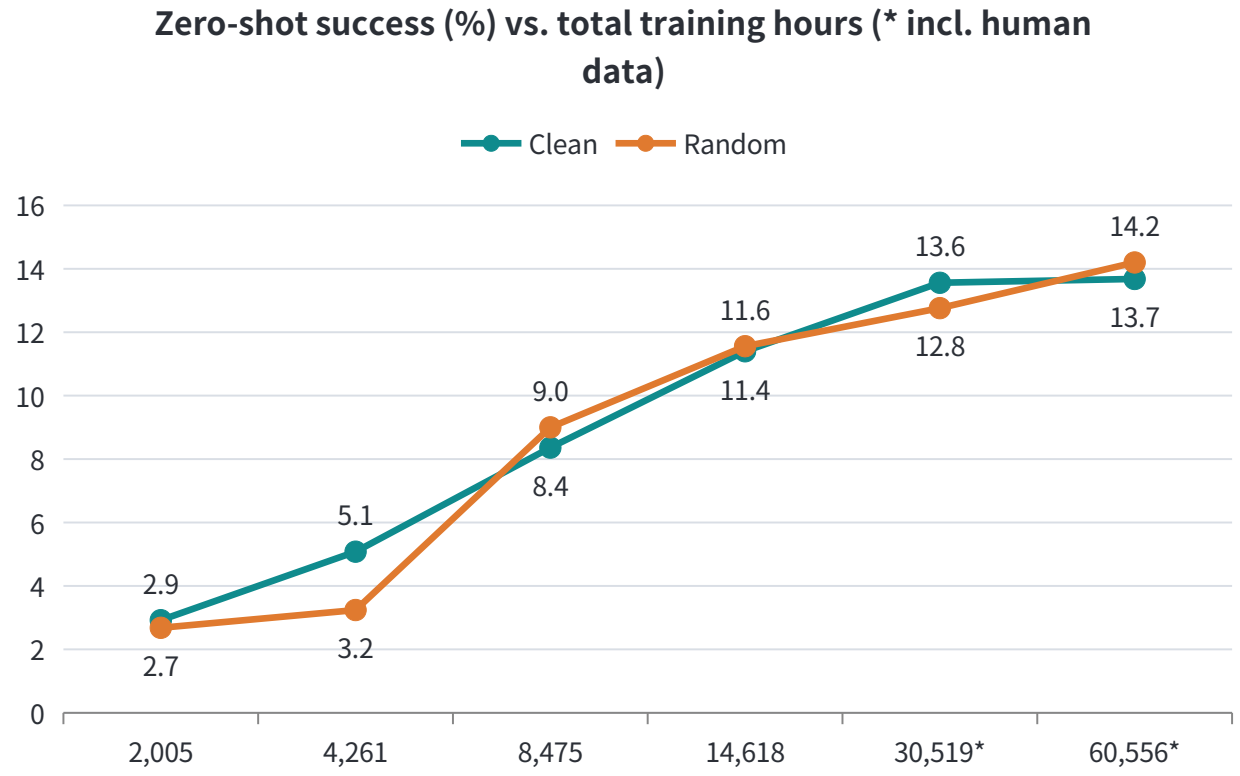
Future 2–8 s frame 40% • final frame 20% • subtask final frame 20% • null goal 20%

Data

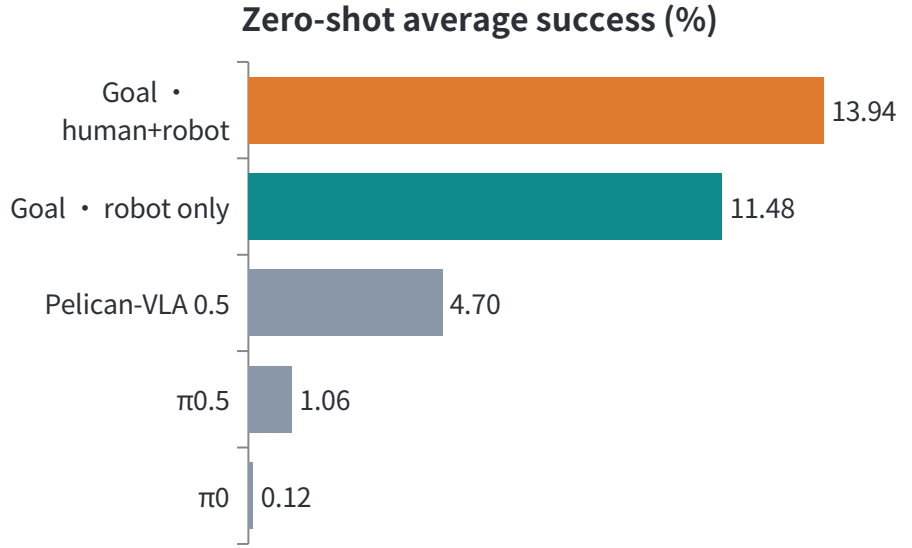
Robot fixed at 14,618 h, plus 15,901 / 45,938 h of human data

ZERO-SHOT ON AN UNSEEN EMBODIMENT

RoboTwin zero-shot: 7.3× robot data drives gains; human data adds 2.46 pp more



First 4 points: robot only; last 2: 14,618 h robot + 15,901 / 45,938 h human data



8 → 18 → 22

Tasks covered per checkpoint (⩾2%); 29 cumulative

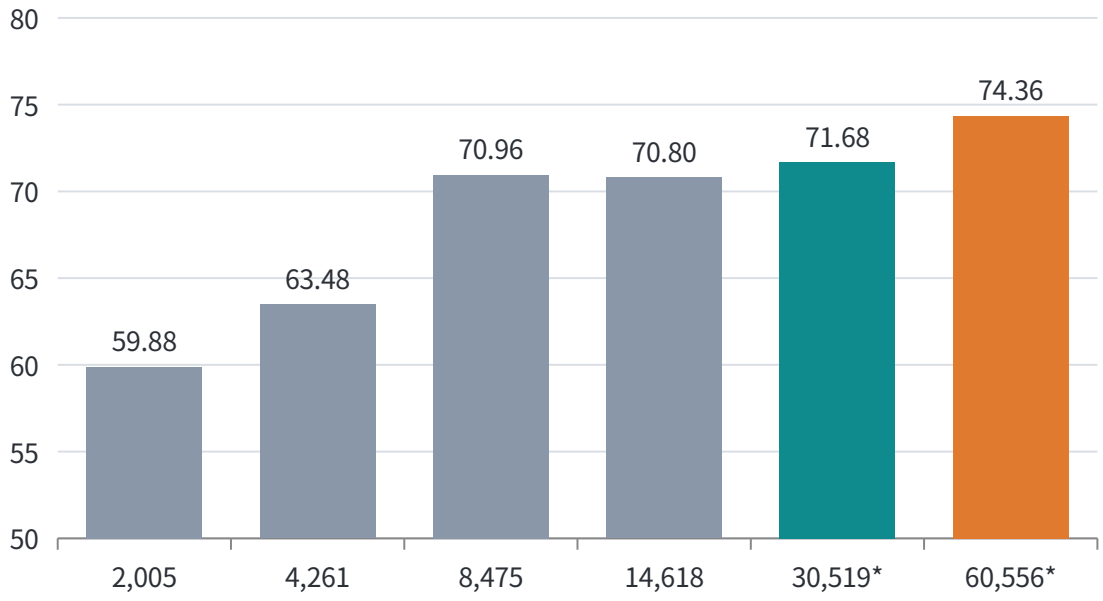
+1.44 vs +0.12

Human data 15.9k→45.9k h: Random vs. Clean gain (pp)

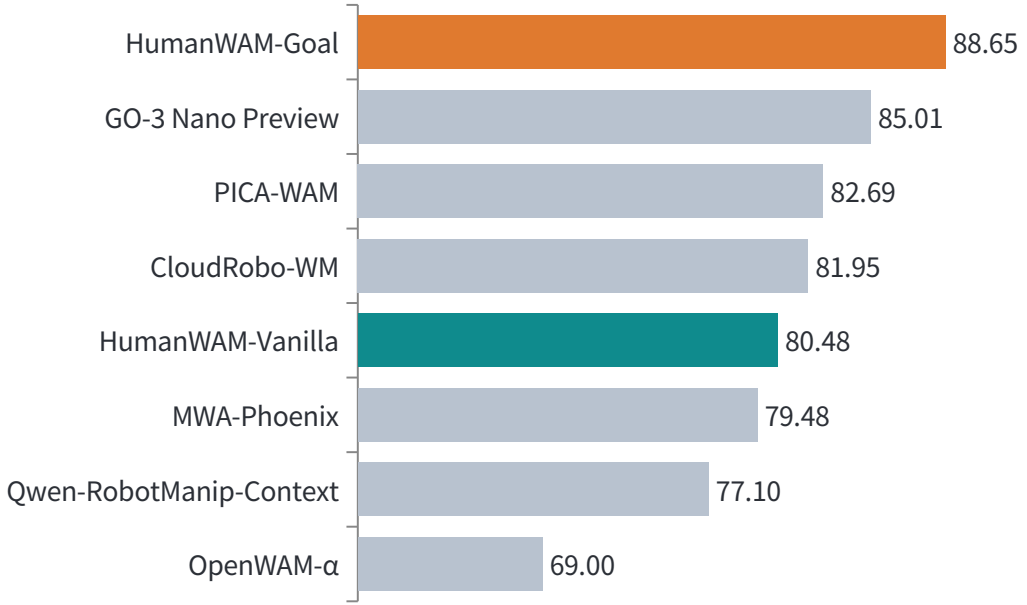
Most open models cannot be evaluated zero-shot: unreleased normalization stats, incompatible control spaces or no public weights (ERR0–2 in Table 13).

Post-train: human data breaks Random plateau; leaderboard 88.65%

Post-training Random success (%) vs. total training hours (* incl. human)



RoboTwin 2.0 leaderboard: (Clean2Clean + Clean2Random)/2



Clean saturates: 87.24 → 89.84 (robot only); 89.00 / 88.96 with human data

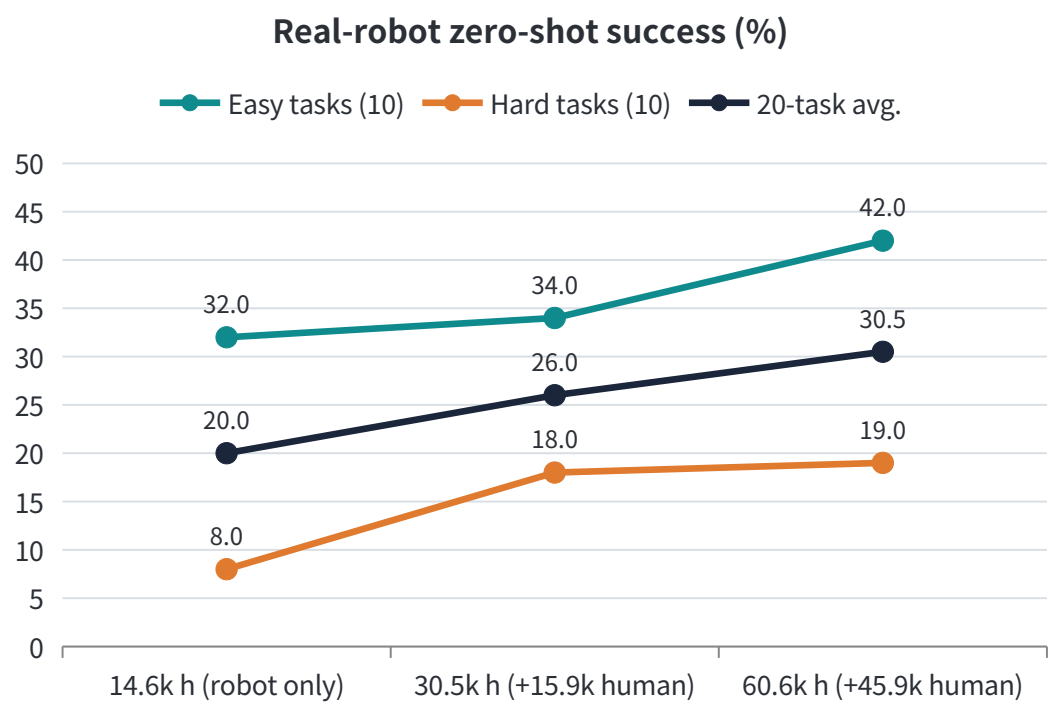
+3.56 pp
Human-data Random gain

92.70 / 84.60
Goal Clean/Random (aug.+new recipe)

+3.64 pp
Avg. tops previous #1, GO-3 Nano Preview

Source: paper Figure 19, Table 14

Real-robot zero-shot: 20-task avg. 20.0% → 30.5%; tasks ever solved: 10 → 16



Task	Success rate (%)				Progress score (0–10)			
	14.6k h Robot only	30.5k h + Human	60.6k h + Human	60.6k h Balanced†	14.6k h Robot only	30.5k h + Human	60.6k h + Human	60.6k h Balanced†
Easy								
Grasp and lift a block	80	70	30	90	8.6	7.6	4.9	9.3
Fully compress a sponge	60	30	70	20	8.0	3.0	7.9	4.3
Open a hardcover book	0	0	0	0	0.0	3.6	2.4	2.2
Ring a bell	60	100	70	20	8.4	10.0	8.5	6.7
Click a ballpoint pen	50	0	50	50	5.0	3.0	5.9	6.5
Open a drawer	0	0	10	0	2.0	3.0	3.4	1.5
Push an object away	0	70	30	20	1.2	7.0	6.3	4.4
Grasp and shake a bottle	60	20	90	100	6.3	1.7	9.0	10.0
Arrange blocks	0	10	20	0	1.3	3.4	3.5	2.7
Place a block on a tray	10	40	50	10	1.3	3.9	4.0	0.8
Hard								
Stack blocks	0	20	30	10	0.9	2.6	4.8	1.6
Place a can beside a book	20	40	20	10	2.5	3.9	3.3	2.9
Place a cup on a coaster	10	10	20	10	1.0	1.0	2.6	1.1
Place an egg in a bowl	30	10	30	30	2.6	1.0	3.3	3.0
Place an object on stacked books	20	40	50	50	1.6	4.0	5.0	5.0
Put a glue stick in a pen holder	0	30	10	40	0.0	3.0	1.0	4.3
Stack two bowls	0	30	0	60	0.0	3.0	1.2	6.0
Lift one bottle with each hand	0	0	30	60	0.0	1.6	6.5	7.8
Open drawer and place object inside	0	0	0	0	0.0	1.0	0.4	0.0
Plug a cable into a power strip	0	0	0	0	0.0	0.0	0.2	1.7
Average · 20 tasks	**20.0**	**26.0**	**30.5**	**29.0**	**2.535**	**3.365**	**4.205**	**4.090**
	0 — 100				0 — 10			

Column headers show total training hours; robot data are fixed at 14,618 h.
10 trials per task. † Balanced: 50% robot samples per batch, with 45,938 h of human data.

10 → 14 → 16

Tasks with ⩾1 success (/20)

19% → 27%

Hard-task success with robot-balanced sampling

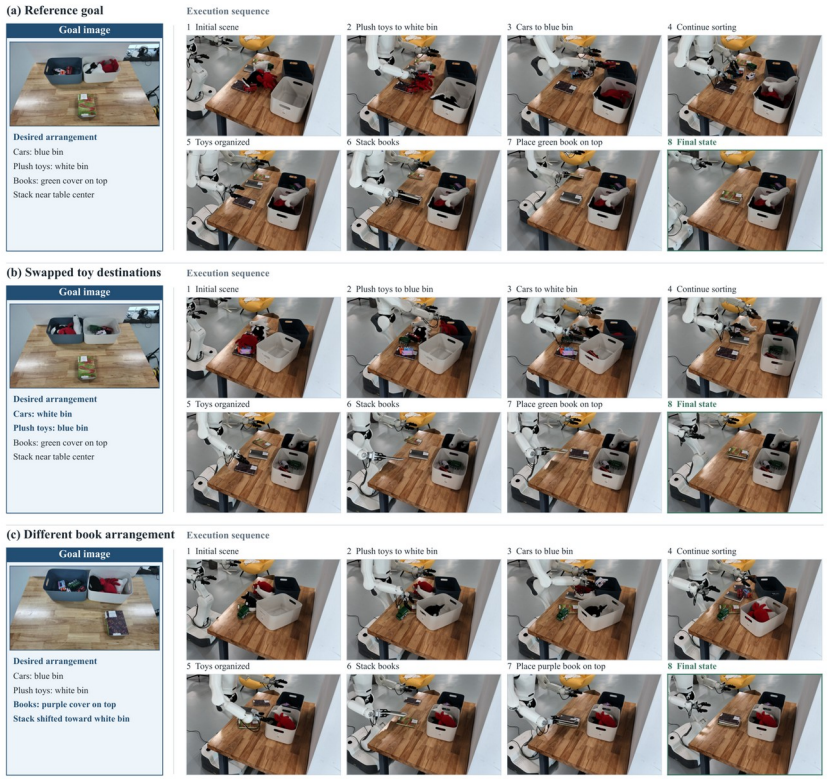
Reading the heatmap

Only 10 trials per task; per-task results non-monotonic (e.g., “pick up block” 80→70→30); balanced 20-task avg. 29.0%, slightly below 30.5%.

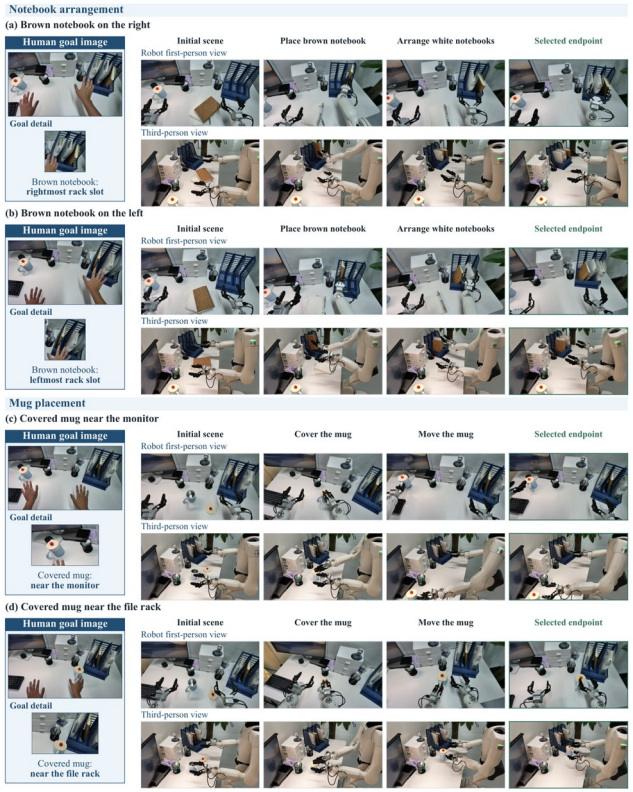
EMBODIED IN-CONTEXT LEARNING

Embodied ICL: frozen model, new goal image → new robot behavior

Post-trained policy • three goals, same scene (Fig. 24)



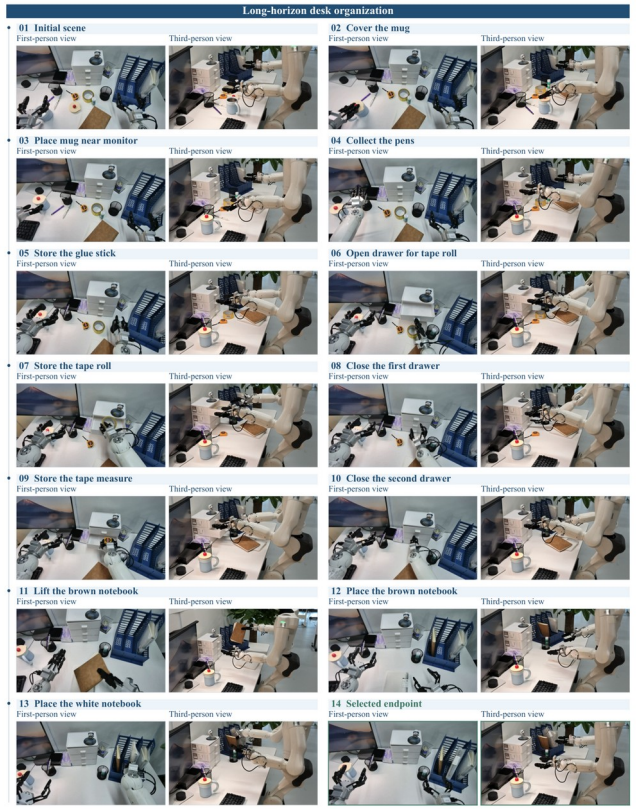
Cross-embodiment • goals from human egocentric photos



Capabilities: changing placement, category→bin mapping, stacking order, drawer choice; the pretrained model (no post-training) already shows early signs (Fig. 23). **Limitations:** all single-rollout qualitative demos, no success-rate statistics.

Long-horizon (post-trained): 14-stage tidying, cross-area chores

14-stage office desk tidying (Fig. 27)



28-keyframe cross-area mobile chores (Fig. 28)

Sort toys | **Stack books** | **Collect bottles at side table** | **Throw in trash** | **Collect clothes from chair** | **Put in laundry basket** | **Return to desk**

CRITICAL ASSESSMENT

Critical view: the direction is credible, but several numbers need discounting

Weak scaling-law statistics

5–7 points per curve, single seed; best checkpoint picked on test set per setting → optimistic numbers; treat as empirical trends

Diversity slope easily misread

Both groups share the 5k h endpoint; at small scale, the broader matched group is higher → the direct conclusion is "broader coverage is better"

SOTA ≠ human only

Goal switches to a 16B backbone, adds robot data and augmentation; RoboTwin goal images synthesized by GPT Image 2.5, not fully comparable to language-only baselines

Noisy real-robot zero-shot

Only 10 trials per task; per-task non-monotonic; robot-balanced overall avg. 29.0% < 30.5%

Low absolute zero-shot rates

≈14% on RoboTwin, ≈30% on real robot; human-data gains are a few pp

Mostly qualitative demos

In-context learning and long-horizon tasks are single rollouts; Vanilla real-robot compared only with No-Ego

TAKEAWAYS

Four actionable takeaways for researchers

01 **Predict human actions + align across sources**

Video-only ≈ no OOD gain; alignment: Random 16.8→26.4%

02 **Broaden coverage, not just hours**

Equal hours: broader task/scene coverage → better results

03 **Measure human data with OOD metrics**

Clean saturates quickly; gains concentrate in Scene-OOD / Random

04 **Preserve robot sample ratio in mixed training**

1:1 sampling lifts hard tasks 19→27%; goals: clearer interface

Open-source release

- Open-HumanWAM-50K corpus (50k hours)
- Data cleaning & action-alignment pipeline
- Pre-/post-training pipelines
- HumanWAM pretrained & post-trained weights
- Inference code & simulation eval suite (with eval seeds)

In one sentence: human-video scale mainly buys robot OOD generalization; mixed with robot data and driven by goal images, zero-shot execution and in-context learning begin to emerge.